

control thread, 1 kHz (mc_rtc FSM + QP)

observe object,
estimate twist,
freeze snapshot s_0

background worker: one frozen snapshot per generation; the control thread polls without blocking

predict $\Pi(h)$
for 14 leads

bank $\mathcal{T} \times \mathcal{G} \times \mathcal{R}$
 $14 \times 32 \times 17$

certify reach, insertion,
closure, retreat
(25 poses per segment)

rank by
 J_{global}

records: every complete plan with its checks, reserves and cost

timing admission
at t_{sel} ; commit-
time consistency

commit once;
execute reach,
capture, retreat